

From Accurate Forecasts to Feasible Plans: Feasibility-Constrained Model Selection for Operational Demand Planning

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Abstract

Forecasting models for operational demand planning are often evaluated by numerical prediction accuracy, yet operational systems do not execute forecasts directly. They execute planning signals such as inventory targets, replenishment recommendations, staffing plans, or capacity allocations. A forecast that improves error metrics can still produce volatile or infeasible plans when downstream execution capacity, model-switching burden, inventory exposure, and planning stability are ignored. This paper formulates forecast-driven demand planning as a feasibility-constrained model selection problem. The proposed evaluation layer converts candidate forecasts into executable planning signals and compares deployable strategies using both raw operational metrics and a normalized planning utility that includes inventory cost, planning volatility, switching burden, and execution violations. Empirically, the draft uses DataCo supply-chain execution data to anchor execution-risk scenarios, Favorita retail demand data as the main forecast-to-plan proof of concept, and M5 retail demand data as a hierarchical robustness setting. The current results suggest that execution-risk-sensitive objectives can reorder deployable strategy rankings, without implying that any single strategy universally dominates. The implication is that operational feasibility should be treated as a first-class planning objective alongside forecast accuracy.

1 Introduction

Demand planning increasingly relies on predictive models to anticipate future demand at fine operational granularity. In many empirical studies and model-selection workflows, the primary criterion for choosing among forecasting methods is numerical prediction accuracy. Forecast accuracy is important because inaccurate forecasts can lead to costly overstocking, stockouts, and service failures. However, forecast accuracy alone does not describe whether the resulting plan can be executed.

In real operations, forecasts are not executed directly. A forecast is converted into a planning signal: an inventory target, a replenishment quantity, a labor plan, a production target, a transportation commitment, or an allocation decision. The operational question is therefore not only whether a forecast is numerically close to realized demand, but whether the planning signal implied by that forecast is stable, feasible, and worth deploying.

A more accurate forecast may create unstable planning signals. It may require frequent model switching, large period-to-period target jumps, repeated infrastructure updates, or adjustment requests that exceed what suppliers, stores, warehouses, transportation teams, and enterprise planning systems can absorb. These burdens may be invisible in forecast-error tables.

This creates a planning-execution gap. Forecasting capability can adapt faster than downstream execution infrastructure. Machine learning pipelines may update daily or weekly, while procurement

lead times, labor scheduling, supplier coordination, production planning, and information-system workflows may change more slowly.

This paper studies forecast-driven demand planning as a feasibility-constrained model selection problem. Rather than asking which forecasting model has the lowest error in isolation, the paper asks which deployable planning strategy balances forecast accuracy, inventory exposure, planning volatility, model-switching burden, and execution-capacity constraints.

The empirical design uses DataCo supply-chain execution data as an execution-risk anchor, Favorita retail demand data as the main forecast-to-plan proof of concept, and M5 retail demand data as a hierarchical robustness setting. Walmart and Rossmann are reserved as optional future robustness extensions.

The paper contributes a feasibility-constrained formulation of forecast-driven demand planning, a normalized planning utility, a DataCo-informed execution-risk scenario design, and empirical evidence that execution-risk-sensitive objectives can reorder deployable strategy rankings across scenarios, planning grains, and demand intermittency.

2 Related Work

2.1 Demand Forecasting and Inventory Implications

Demand forecasting research studies how predictive models can estimate future demand across products, stores, regions, and time periods. A closely related stream connects forecast quality to downstream inventory and service outcomes. This paper builds on that operational motivation but focuses on the conversion of candidate forecasts into planning signals that must be executed under finite adjustment capacity.

2.2 Forecast Combination and Model Selection

Forecast-combination and model-selection methods provide mechanisms for choosing among or aggregating candidate forecasts. This paper differs from standard forecast-combination work because the selection criterion is deployable planning utility rather than forecast error alone.

2.3 Cost-Sensitive and Multi-Objective Decision Making

Cost-sensitive and multi-objective decision methods are related because demand planning rarely has a single objective. The contribution here is a structured operational planning evaluation framework in which forecast-driven plans are compared against execution-capacity constraints and normalized relative to an accuracy-first baseline.

2.4 Supply-Chain Execution Risk and Planning Stability

Supply-chain execution risk, planning nervousness, and plan volatility are central to the gap studied in this paper. This paper connects forecast-driven model selection to measurable execution feasibility by using supply-chain execution evidence to define scenario weights and by evaluating planning volatility and execution violations directly.

3 Problem Formulation

Let $i \in \mathcal{I}$ index a planning unit and let $t \in \mathcal{T}$ index a planning period. Let $m \in \mathcal{M}$ denote a candidate forecasting model or forecasting strategy. Actual demand is $y_{i,t}$, and the forecast

produced by model m is $\hat{y}_{i,t}^m$. A deployable decision strategy is denoted by $s \in \mathcal{S}$.

The forecasting layer produces candidate forecasts $\{\hat{y}_{i,t}^m : m \in \mathcal{M}\}$. A hard-selection strategy chooses one model $z_{i,t}^s$ and sets

$$\tilde{y}_{i,t}^s = \hat{y}_{i,t}^{z_{i,t}^s}. \quad (1)$$

A soft-composition strategy assigns nonnegative weights $w_{i,t}^{m,s}$ satisfying $\sum_{m \in \mathcal{M}} w_{i,t}^{m,s} = 1$ and sets

$$\tilde{y}_{i,t}^s = \sum_{m \in \mathcal{M}} w_{i,t}^{m,s} \hat{y}_{i,t}^m. \quad (2)$$

The operation executes a planning signal rather than a raw forecast. Let policy_t denote the planning policy in effect at time t . The forecast-to-plan conversion is

$$x_{i,t}^s = g(\tilde{y}_{i,t}^s, \text{policy}_t). \quad (3)$$

The running implementation uses an inventory target as the planning signal.

Feasibility-aware smoothing changes the executed planning signal, not the realized demand and not the raw candidate forecast:

$$x_{i,t}^{\text{exec}} = \alpha_t x_{i,t}^{\text{cand}} + (1 - \alpha_t) x_{i,t-1}^{\text{exec}}. \quad (4)$$

Inventory cost includes holding and shortage costs:

$$\text{holding}_{i,t}^s = h \max(x_{i,t}^s - y_{i,t}, 0), \quad (5)$$

$$\text{shortage}_{i,t}^s = p \max(y_{i,t} - x_{i,t}^s, 0), \quad (6)$$

$$\text{inventory_cost}_{i,t}^s = \text{holding}_{i,t}^s + \text{shortage}_{i,t}^s. \quad (7)$$

Planning volatility is measured by the period-to-period change in the executable signal, and execution violations are measured as

$$\text{execution_violation}_{i,t}^s = \max(0, |x_{i,t}^s - x_{i,t-1}^s| - C_{i,t}), \quad (8)$$

where $C_{i,t}$ is execution capacity.

For scalar ranking, the analysis reports normalized planning loss:

$$\begin{aligned} \text{normalized_total_loss}^s = & \frac{\text{inventory_cost}^s}{\text{inventory_cost}_{\text{ref}}} + \lambda_{\text{volatility}} \frac{\text{planning_volatility}^s}{\text{volatility}_{\text{ref}}} \\ & + \lambda_{\text{execution}} \frac{\text{execution_penalty}^s}{\text{execution_penalty}_{\text{ref}}} + \lambda_{\text{switch}} \frac{\text{switching_burden}^s}{\text{switching_burden}_{\text{ref}}}. \end{aligned} \quad (9)$$

The reference strategy is the accuracy-first BestAccuracy or Global Best baseline within the same dataset, run mode, and comparison split. DataCo-informed $\lambda_{\text{execution}}$ values are sensitivity weights, not exact economic cost calibrations.

The deployable finite-horizon selector minimizes cumulative expected operational loss using validation-derived expected costs:

$$\min_{z_{i,1:T}} \sum_{t=1}^T \ell_{i,t} \left(z_{i,t}, z_{i,t-1}, x_{i,t}^{z_{i,t}}, x_{i,t-1}^{z_{i,t-1}} \right). \quad (10)$$

A budgeted version adds $\sum_{t=2}^T \mathbb{I}\{z_{i,t} \neq z_{i,t-1}\} \leq K$. Realized test demand is not used for deployable DP selection.

4 Framework and Methods

The framework separates forecast generation from deployable planning strategy selection. Candidate forecasting models first produce forecasts. A decision layer then selects, combines, or smooths the candidate signals before converting them into executable planning signals.

Accuracy-first methods include Global Best or BestAccuracy and Family Best. Stability-first methods include moving-average and BestStability variants. Ensemble-based methods include Simple Ensemble and Operational-Loss Ensemble. Smoothing-based methods include FeasibilityAwareSmoothed with fixed alpha and utility-CV alpha. Feasibility-aware adaptive selection evaluates candidate switches using expected planning utility rather than forecast error alone. The finite-horizon decision layer includes GreedyFeasibilitySelector, DPFeasibilitySelector, and BudgetedDPFeasibilitySelector. Oracle uses realized demand information and is non-deployable; it is included as a diagnostic reference.

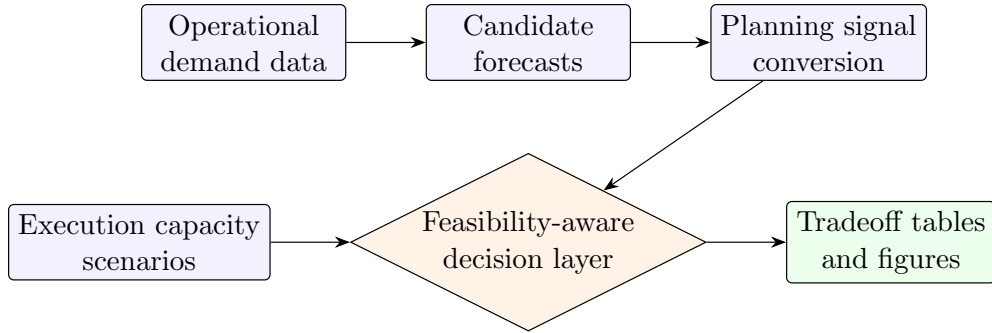


Figure 1: Feasibility-aware demand-planning analysis workflow.

5 Experimental Setup

DataCo is used as an execution-risk anchor. Favorita is the main forecast-to-plan proof-of-concept dataset, with demand, promotion, holiday, oil-price, transaction, and store-context signals. M5 is used as a large-scale hierarchical robustness dataset across item, department, category, store, and state levels. Walmart and Rossmann are reserved as optional future robustness extensions.

The Favorita analysis includes transparent baselines and machine-learning forecasters, including naive, seasonal, moving-average, exponential-smoothing, XGBoost, and LightGBM variants where available. The M5 robustness experiments use transparent forecast candidates that are computationally feasible for hierarchical sensitivity checks.

The evaluation reports WAPE, inventory cost, planning volatility, execution penalty, execution violation rate, model-switching burden, normalized total loss, and gap to the Oracle diagnostic. DataCo-derived late-delivery-risk percentiles are mapped into execution-risk sensitivity scenarios using a configurable formula.

6 Results

6.1 Execution Feasibility Is Empirically Measurable

The DataCo analysis reports a global late-delivery rate of 0.548, with context-level late-delivery-rate percentiles of 0.535 at the 30th percentile, 0.552 at the median, 0.581 at the 75th percentile,

Table 1: DataCo late-delivery risk percentiles used as execution-risk anchors.

percentile_name	percentile	late_delivery_rate	source	fallback_used	fallback_reason
p30_late_delivery_rate	30	0.535	dataco_derived	False	
p50_late_delivery_rate	50	0.552	dataco_derived	False	
p75_late_delivery_rate	75	0.581	dataco_derived	False	
p95_late_delivery_rate	95	0.901	dataco_derived	False	

Table 2: Generated execution-risk scenarios for DataCo-informed re-evaluation.

scenario_name	percentile_anchor	late_delivery_rate_anchor	lambda_execution	mapping_formula	source	fallback_used
baseline	0	0.000	0.050	$\lambda_{\text{execution}} = \text{clip}(\text{base_lambda} + \text{risk_sen...})$	dataco_derived	False
dataco_low	30	0.535	0.318	$\lambda_{\text{execution}} = \text{clip}(\text{base_lambda} + \text{risk_sen...})$	dataco_derived	False
dataco_median	50	0.552	0.326	$\lambda_{\text{execution}} = \text{clip}(\text{base_lambda} + \text{risk_sen...})$	dataco_derived	False
dataco_high	75	0.581	0.341	$\lambda_{\text{execution}} = \text{clip}(\text{base_lambda} + \text{risk_sen...})$	dataco_derived	False
dataco_severe	95	0.901	0.500	$\lambda_{\text{execution}} = \text{clip}(\text{base_lambda} + \text{risk_sen...})$	dataco_derived	False

and 0.901 at the 95th percentile. These values anchor execution-risk sensitivity scenarios rather than direct economic cost calibration.

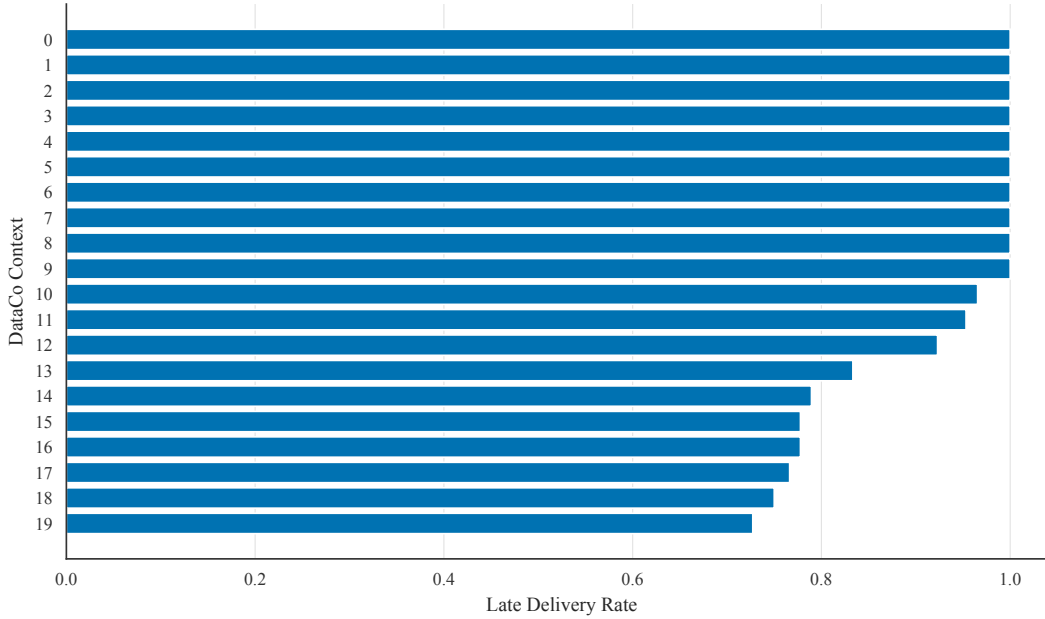


Figure 2: DataCo late-delivery risk across operational contexts.

6.2 Accuracy-First Methods Remain Strong Forecasters but Can Create Execution Burden

In Favorita, LightGBM has the lowest test WAPE among evaluated forecasting models, with WAPE 0.109. The accuracy-first Global Best planning strategy has WAPE 0.109, inventory cost 3,010,925, execution penalty 546,133, execution violation rate 0.284, and normalized total loss 1.550 in the baseline method-family table.

Table 3: Favorita forecast accuracy metrics from the minimal pipeline.

Split	Model	Rows	MAE	RMSE	WAPE
test	LightGBM	2800	492.761	725.326	0.109
test	XGBoost	2800	502.155	739.079	0.111
test	Seasonal	2800	713.025	1065.398	0.157
test	Moving Avg.	2800	852.546	1198.148	0.188
test	Exp. Smooth	2800	881.606	1244.260	0.194
test	Naive	2800	942.064	1378.074	0.208
validation	LightGBM	2800	442.962	684.316	0.095
validation	XGBoost	2800	447.151	702.795	0.096
validation	Seasonal	2800	729.933	1133.405	0.157
validation	Moving Avg.	2800	1076.697	1554.387	0.231
validation	Naive	2800	1146.079	1718.333	0.246
validation	Exp. Smooth	2800	1147.294	1651.643	0.247

6.3 DataCo-Informed Execution Scenarios Reorder Deployable Strategy Rankings

In the severe DataCo-informed Favorita scenario, Simple Ensemble has normalized total loss 1.473 and Operational Ensemble has normalized total loss 1.496, while Global Best has normalized total loss 2.000. This indicates that execution-risk-sensitive objectives can reorder deployable strategy rankings, without implying that any ensemble strategy universally dominates.

6.4 Smoothing and Ensemble Methods Occupy Different Frontier Regions

Smoothed Alpha 0.25 has WAPE 0.109 in the baseline Favorita scenario and reduces execution penalty to 102, but inventory cost rises to 4,515,961. Operational Ensemble has a more moderate accuracy penalty than Simple Ensemble and substantially lower execution penalty than Global Best. These patterns support a tradeoff interpretation rather than a universal-winner interpretation.

6.5 M5 Confirms Robustness in Hierarchical Demand Planning

In M5 quick-mode item-store replication, Simple Ensemble has the lowest non-oracle WAPE at 0.332, but execution penalty 4,451 and violation rate 0.233. Smoothed Alpha 0.25 has WAPE 0.368, execution penalty 101, violation rate 0.010, and normalized total loss 1.223. Operational Ensemble has WAPE 0.356, execution penalty 302, violation rate 0.024, and normalized total loss 1.531.

Table 4: Representative Favorita method-family comparison under DataCo-informed execution scenarios.

run_mode	scenario_name	method_family	method_name	strategy	WAPPE	inventory_cost	planning_latency	execution_penalty	max_penalty	execution_violation_rate	model_switch_count	max_penalty_plan_change_rate	normalized_total_cost	gap_ratio
quick	baseline	Oracle	Oracle	oracle_realized_demand	0.000	1638182.571	487.867	680523.022	0.345	0	1.351	1.193	0.0000	
quick	baseline	FeasibilityAwareSmoothed	Operational Ensemble	feasibility_aware_smoothed.inverse_operational...	0.119	3201550.617	288.690	118441.378	0.133	0	0.615	1.422	0.229	
quick	baseline	ReferenceEnsemble	Simple Ensemble	simple_ensemble	0.120	312038.615	246.143	4721.135	0.066	0	0.578	1.434	0.242	
quick	baseline	AccuracyFirst	Family Best	family_best_ensemble	0.119	301092.1926	411.964	540133.345	0.284	0	0.764	1.563	0.370	
quick	baseline	AccuracyFirst	Family Best	family_best_ensemble	0.110	3064402.341	411.535	537406.724	0.279	0	0.764	1.563	0.370	
quick	baseline	FeasibilityAwareSmoothed	Smoothed Alpha 0.25	feasibility_aware_smoothed.alpha.0.25	0.109	4151961.102	109.008	102.215	0.001	0	0.726	1.646	0.454	
quick	baseline	FeasibilityAwareSmoothed	Stability First	best_stability_ensemble	0.110	3203645.540	109.008	102.215	0.001	0	0.726	1.646	0.454	
quick	baseline	FeasibilityAwareSmoothed	Utility Alpha	feasibility_aware_smoothed.utility.alpha	0.109	5373551.068	46.201	0.000	0.000	0	0.095	1.836	0.643	
quick	baseline	FeasibilityAwareSelector	Feasibility-Aware	feasibility_aware_selector	0.116	3757551.068	364.328	305116.914	0.232	781	0.581	2.911	1.718	
quick	baseline_high	ReferenceEnsemble	Simple Ensemble	simple_ensemble	0.120	32038.615	246.143	4721.135	0.066	0	0.578	1.434	0.242	
quick	data_low_high	FeasibilityAwareEnsemble	Operational Ensemble	feasibility_aware_ensemble.inverse_operational...	0.120	322771.072	281.189	102559.642	0.122	0	0.598	1.473	-0.081	
quick	data_low_high	FeasibilityAwareSmoothed	Operational Alpha 0.25	oracle_realized_demand	0.119	363181.571	487.867	680523.022	0.345	0	1.351	1.555	0.000	
quick	data_low_high	FeasibilityAwareSmoothed	Stability First	feasibility_aware_smoothed.alpha.0.25	0.109	4151961.102	102.215	0.000	0.001	0	0.726	1.646	0.454	
quick	data_low_high	FeasibilityAwareSmoothed	Best Stability	best_stability_ensemble	0.188	5263645.540	14.814	0.000	0.000	0	0.128	1.706	0.221	
quick	data_low_high	FeasibilityAwareSmoothed	Utility Alpha	feasibility_aware_smoothed.utility.alpha	0.109	5373682.296	42.201	0.000	0.000	0	0.095	1.836	0.281	
quick	data_low_high	AccuracyFirst	Family Best	family_best_ensemble	0.110	301092.1926	411.964	540133.345	0.284	0	0.764	1.563	0.370	
quick	data_low_high	AccuracyFirst	Family Best	family_best_ensemble	0.110	3064402.341	411.535	537406.724	0.279	0	0.764	1.563	0.370	
quick	data_low_high	FeasibilityAwareSelector	Feasibility-Aware	feasibility_aware_selector	0.116	3757551.068	364.328	305116.914	0.232	781	0.581	2.911	1.718	
quick	data_low_high	ReferenceEnsemble	Simple Ensemble	simple_ensemble	0.120	32038.615	246.143	4721.135	0.066	0	0.578	1.434	0.242	
quick	data_low_low	FeasibilityAwareEnsemble	Operational Ensemble	feasibility_aware_ensemble.inverse_operational...	0.120	322973.229	281.708	104381.142	0.122	0	0.599	1.470	-0.056	
quick	data_low_low	FeasibilityAwareSmoothed	Operational Alpha 0.25	oracle_realized_demand	0.119	363181.571	487.867	680523.022	0.345	0	1.351	1.526	0.000	
quick	data_low_low	FeasibilityAwareSmoothed	Smoothed Alpha 0.25	feasibility_aware_smoothed.alpha.0.25	0.109	4151961.102	109.008	102.215	0.001	0	0.726	1.655	0.129	
quick	data_low_low	FeasibilityAwareSmoothed	Stability First	best_stability_ensemble	0.188	5263645.540	14.814	0.000	0.000	0	0.128	1.740	0.240	
quick	data_low_low	AccuracyFirst	Family Best	family_best_ensemble	0.110	301092.1926	411.964	540133.345	0.284	0	0.764	1.518	0.027	
quick	data_low_low	AccuracyFirst	Family Best	family_best_ensemble	0.110	3064402.341	411.535	537406.724	0.279	0	0.764	1.526	0.030	
quick	data_low_low	FeasibilityAwareSmoothed	Utility Alpha	feasibility_aware_smoothed.utility.alpha	0.109	5373682.296	42.201	0.000	0.000	0	0.095	1.836	0.281	
quick	data_low_low	FeasibilityAwareSelector	Feasibility-Aware	feasibility_aware_selector	0.116	3757551.068	364.328	305116.914	0.232	781	0.581	2.861	2.335	

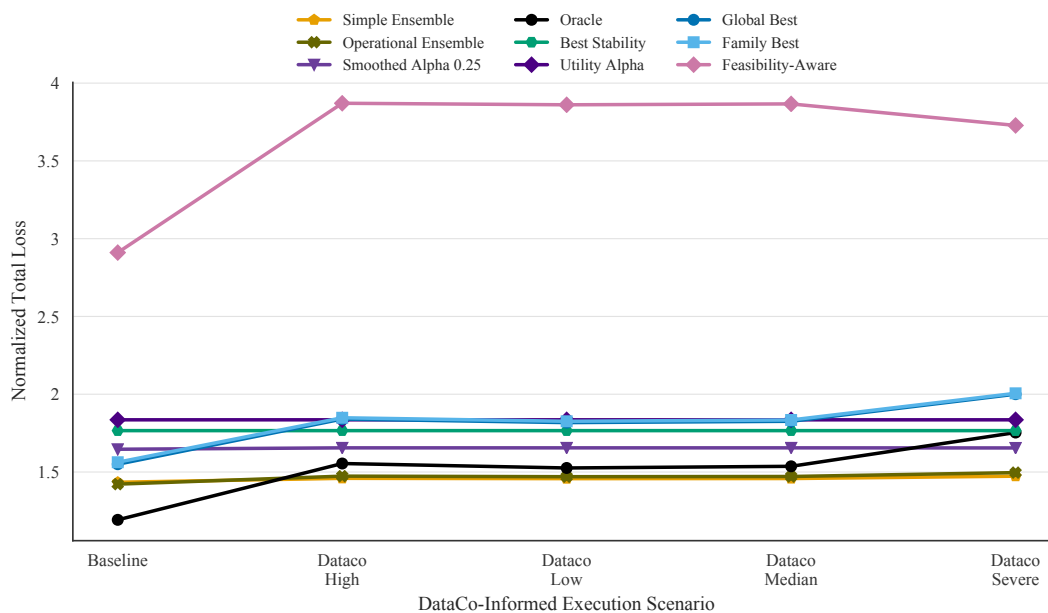


Figure 3: Favorita normalized planning loss across DataCo-informed execution-risk scenarios.

Table 5: M5 large-scale replication of the accuracy-feasibility planning comparison.

Table 1: Performance of the proposed algorithm on various datasets									
Model	Architecture	Ways	Number of nodes	Number of edges	Number of clusters	Number of clusters per node	Number of clusters per edge	Number of clusters per node per edge	Number of clusters per node per edge per cluster
1	graph	graph	40	1000	1000	1000	1000	1000	1000
2	graph	graph	40	1000	1000	1000	1000	1000	1000
3	graph	graph	40	1000	1000	1000	1000	1000	1000
4	graph	graph	40	1000	1000	1000	1000	1000	1000
5	graph	graph	40	1000	1000	1000	1000	1000	1000
6	graph	graph	40	1000	1000	1000	1000	1000	1000
7	graph	graph	40	1000	1000	1000	1000	1000	1000
8	graph	graph	40	1000	1000	1000	1000	1000	1000
9	graph	graph	40	1000	1000	1000	1000	1000	1000
10	graph	graph	40	1000	1000	1000	1000	1000	1000
11	graph	graph	40	1000	1000	1000	1000	1000	1000
12	graph	graph	40	1000	1000	1000	1000	1000	1000
13	graph	graph	40	1000	1000	1000	1000	1000	1000
14	graph	graph	40	1000	1000	1000	1000	1000	1000
15	graph	graph	40	1000	1000	1000	1000	1000	1000
16	graph	graph	40	1000	1000	1000	1000	1000	1000
17	graph	graph	40	1000	1000	1000	1000	1000	1000
18	graph	graph	40	1000	1000	1000	1000	1000	1000
19	graph	graph	40	1000	1000	1000	1000	1000	1000
20	graph	graph	40	1000	1000	1000	1000	1000	1000
21	graph	graph	40	1000	1000	1000	1000	1000	1000
22	graph	graph	40	1000	1000	1000	1000	1000	1000
23	graph	graph	40	1000	1000	1000	1000	1000	1000
24	graph	graph	40	1000	1000	1000	1000	1000	1000
25	graph	graph	40	1000	1000	1000	1000	1000	1000
26	graph	graph	40	1000	1000	1000	1000	1000	1000
27	graph	graph	40	1000	1000	1000	1000	1000	1000
28	graph	graph	40	1000	1000	1000	1000	1000	1000
29	graph	graph	40	1000	1000	1000	1000	1000	1000
30	graph	graph	40	1000	1000	1000	1000	1000	1000
31	graph	graph	40	1000	1000	1000	1000	1000	1000
32	graph	graph	40	1000	1000	1000	1000	1000	1000
33	graph	graph	40	1000	1000	1000	1000	1000	1000
34	graph	graph	40	1000	1000	1000	1000	1000	1000
35	graph	graph	40	1000	1000	1000	1000	1000	1000
36	graph	graph	40	1000	1000	1000	1000	1000	1000
37	graph	graph	40	1000	1000	1000	1000	1000	1000
38	graph	graph	40	1000	1000	1000	1000	1000	1000
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41	graph	graph	40	1000	1000	1000	1000	1000	1000
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43	graph	graph	40	1000	1000	1000	1000	1000	1000
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97	graph	graph	40	1000	1000	1000	1000	1000	1000
98	graph	graph	40	1000	1000	1000	1000	1000	1000
99	graph	graph	40	1000	1000	1000	1000	1000	1000
100	graph	graph	40	1000	1000	1000	1000	1000	1000

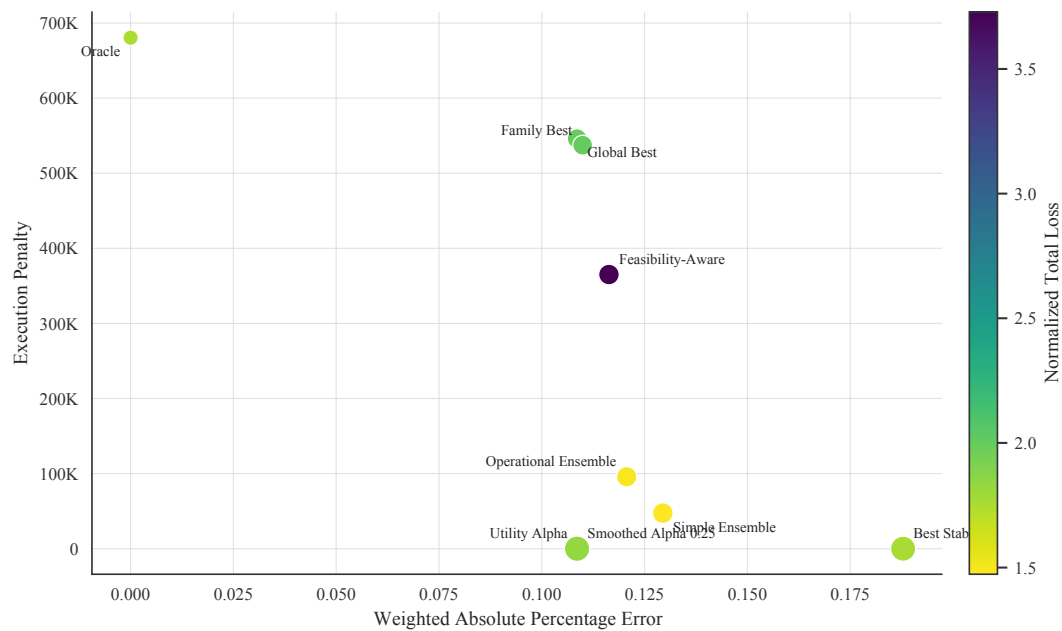


Figure 4: Favorita method-family frontier between forecast accuracy and execution penalty.

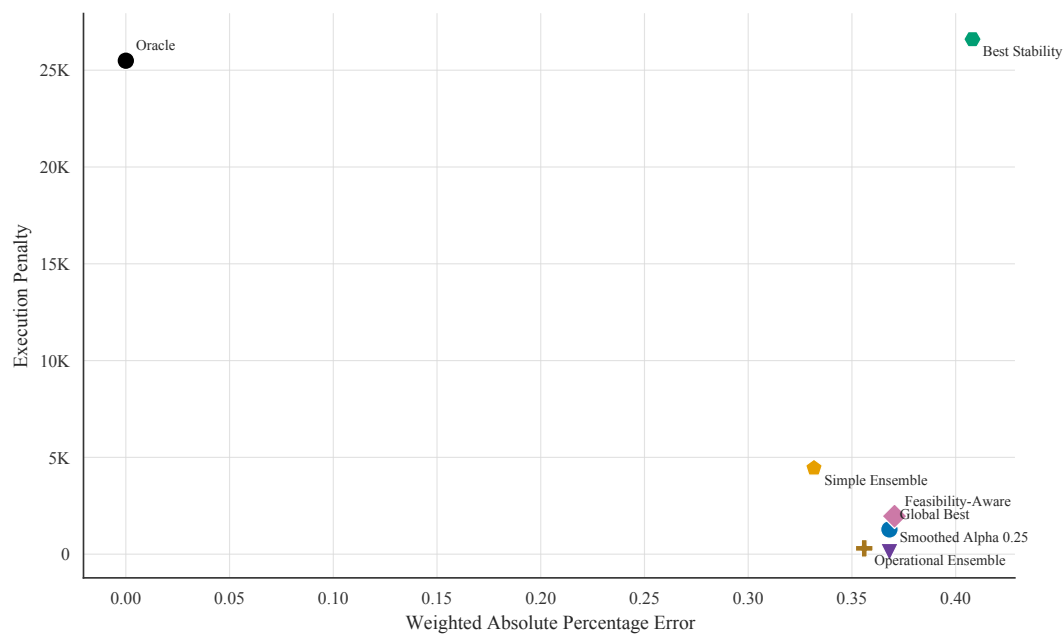


Figure 5: M5 accuracy versus execution penalty across deployable strategy families.

[illegible]

	Region	Country	City	Lat	Long	Alt	Area	Pop	PopD	PopD2	PopD3	PopD4	PopD5	PopD6	PopD7	PopD8	PopD9	PopD10	PopD11	PopD12	PopD13	PopD14	PopD15	PopD16	PopD17	PopD18	PopD19	PopD20	PopD21	PopD22	PopD23	PopD24	PopD25	PopD26	PopD27	PopD28	PopD29	PopD30	PopD31	PopD32	PopD33	PopD34	PopD35	PopD36	PopD37	PopD38	PopD39	PopD40	PopD41	PopD42	PopD43	PopD44	PopD45	PopD46	PopD47	PopD48	PopD49	PopD50	PopD51	PopD52	PopD53	PopD54	PopD55	PopD56	PopD57	PopD58	PopD59	PopD60	PopD61	PopD62	PopD63	PopD64	PopD65	PopD66	PopD67	PopD68	PopD69	PopD70	PopD71	PopD72	PopD73	PopD74	PopD75	PopD76	PopD77	PopD78	PopD79	PopD80	PopD81	PopD82	PopD83	PopD84	PopD85	PopD86	PopD87	PopD88	PopD89	PopD90	PopD91	PopD92	PopD93	PopD94	PopD95	PopD96	PopD97	PopD98	PopD99	PopD100	PopD101	PopD102	PopD103	PopD104	PopD105	PopD106	PopD107	PopD108	PopD109	PopD110	PopD111	PopD112	PopD113	PopD114	PopD115	PopD116	PopD117	PopD118	PopD119	PopD120	PopD121	PopD122	PopD123	PopD124	PopD125	PopD126	PopD127	PopD128	PopD129	PopD130	PopD131	PopD132	PopD133	PopD134	PopD135	PopD136	PopD137	PopD138	PopD139	PopD140	PopD141	PopD142	PopD143	PopD144	PopD145	PopD146	PopD147	PopD148	PopD149	PopD150	PopD151	PopD152	PopD153	PopD154	PopD155	PopD156	PopD157	PopD158	PopD159	PopD160	PopD161	PopD162	PopD163	PopD164	PopD165	PopD166	PopD167	PopD168	PopD169	PopD170	PopD171	PopD172	PopD173	PopD174	PopD175	PopD176	PopD177	PopD178	PopD179	PopD180	PopD181	PopD182	PopD183	PopD184	PopD185	PopD186	PopD187	PopD188	PopD189	PopD190	PopD191	PopD192	PopD193	PopD194	PopD195	PopD196	PopD197	PopD198	PopD199	PopD200	PopD201	PopD202	PopD203	PopD204	PopD205	PopD206	PopD207	PopD208	PopD209	PopD210	PopD211	PopD212	PopD213	PopD214	PopD215	PopD216	PopD217	PopD218	PopD219	PopD220	PopD221	PopD222	PopD223	PopD224	PopD225	PopD226	PopD227	PopD228	PopD229	PopD230	PopD231	PopD232	PopD233	PopD234	PopD235	PopD236	PopD237	PopD238	PopD239	PopD240	PopD241	PopD242	PopD243	PopD244	PopD245	PopD246	PopD247	PopD248	PopD249	PopD250	PopD251	PopD252	PopD253	PopD254	PopD255	PopD256	PopD257	PopD258	PopD259	PopD260	PopD261	PopD262	PopD263	PopD264	PopD265	PopD266	PopD267	PopD268	PopD269	PopD270	PopD271	PopD272	PopD273	PopD274	PopD275	PopD276	PopD277	PopD278	PopD279	PopD280	PopD281	PopD282	PopD283	PopD284	PopD285	PopD286	PopD287	PopD288	PopD289	PopD290	PopD291	PopD292	PopD293	PopD294	PopD295	PopD296	PopD297	PopD298	PopD299	PopD300	PopD301	PopD302	PopD303	PopD304	PopD305	PopD306	PopD307	PopD308	PopD309	PopD310	PopD311	PopD312	PopD313	PopD314	PopD315	PopD316	PopD317	PopD318	PopD319	PopD320	PopD321	PopD322	PopD323	PopD324	PopD325	PopD326	PopD327	PopD328	PopD329	PopD330	PopD331	PopD332	PopD333	PopD334	PopD335	PopD336	PopD337	PopD338	PopD339	PopD340	PopD341	PopD342	PopD343	PopD344	PopD345	PopD346	PopD347	PopD348	PopD349	PopD350	PopD351	PopD352	PopD353	PopD354	PopD355	PopD356	PopD357	PopD358	PopD359	PopD360	PopD361	PopD362	PopD363	PopD364	PopD365	PopD366	PopD367	PopD368	PopD369	PopD370	PopD371	PopD372	PopD373	PopD374	PopD375	PopD376	PopD377	PopD378	PopD379	PopD380	PopD381	PopD382	PopD383	PopD384	PopD385	PopD386	PopD387	PopD388	PopD389	PopD390	PopD391	PopD392	PopD393	PopD394	PopD395	PopD396	PopD397	PopD398	PopD399	PopD400	PopD401	PopD402	PopD403	PopD404	PopD405	PopD406	PopD407	PopD408	PopD409	PopD410	PopD411	PopD412	PopD413	PopD414
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This line chart illustrates the Normalized Total Loss (Y-axis, ranging from 0 to 25) as a function of the Planning Grain (X-axis, with categories: Item Store, Department Store, and Category Store). The chart compares several methods:

- Global Best** (Blue solid line with circles)
- Simple Ensemble** (Yellow solid line with circles)
- Operational Ensemble** (Brown dashed line with circles)
- Smoothed Alpha 0.25** (Purple solid line with circles)
- Feasibility-Aware** (Pink solid line with diamonds)
- Best Stability** (Teal solid line with circles)
- Oracle** (Black dotted line with circles)

The Oracle method shows the highest loss at the Item Store grain (approx. 23) and decreases significantly as the planning grain increases. The Best Stability method follows a similar trend but with slightly lower loss than the Oracle. The Feasibility-Aware method starts with a loss of approx. 5.5 at the Item Store grain and decreases to approx. 2.5 at the Category Store grain. The Global Best, Simple Ensemble, and Operational Ensemble methods show much lower losses at the Item Store grain (approx. 2.5, 5.0, and 1.5 respectively) and remain relatively stable or slightly decrease as the planning grain increases.

Method	Item Store	Department Store	Category Store
Global Best	2.5	2.5	2.5
Simple Ensemble	5.0	1.5	1.5
Operational Ensemble	1.5	1.5	1.5
Smoothed Alpha 0.25	1.5	1.5	1.5
Feasibility-Aware	5.5	2.5	2.5
Best Stability	22.5	2.5	2.5
Oracle	23.0	2.5	2.5

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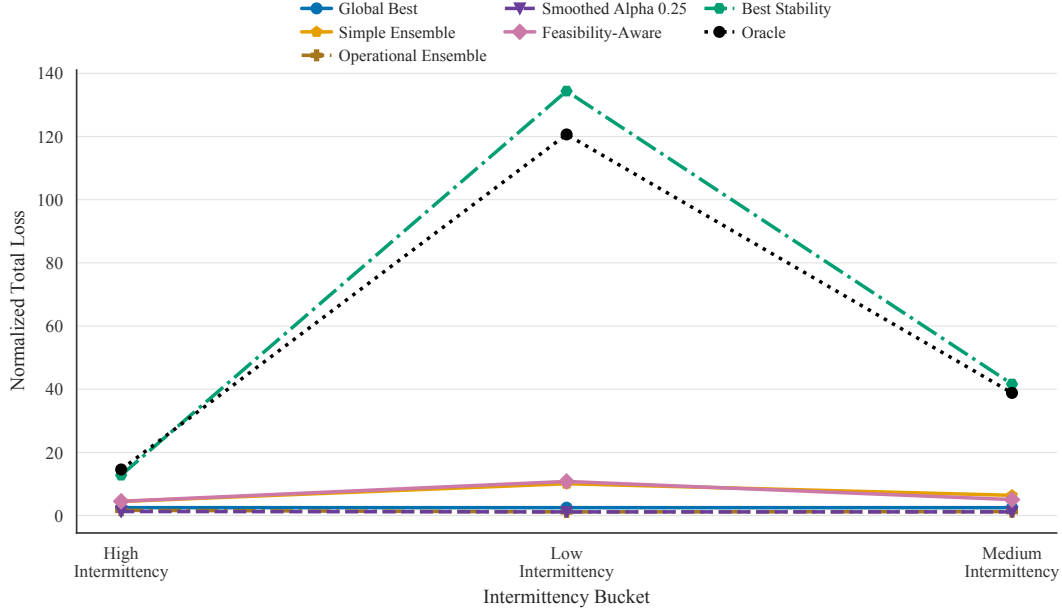


Figure 7: M5 intermittent-demand stress test.

7 Discussion

The empirical results should not be read as evidence that simple models beat machine-learning models. The central point is that once forecasts are converted into plans, deployability depends on additional objectives that forecast-error metrics do not observe. The results also do not imply that smoothing solves every planning problem. Smoothing reduces abrupt plan changes by construction, but it can increase inventory exposure when the executed plan adapts too slowly.

Operational feasibility should be treated as a first-class evaluation objective because planning signals create work for the execution system. DataCo is used as an empirical execution-risk anchor, not as direct economic cost calibration for Favorita or M5. The practical implication is that model deployment should be evaluated at the forecast-to-plan layer, with explicit attention to execution capacity and planning granularity.

Limitations include the absence of full internal execution-cost observations in public datasets, proxied execution capacity in Favorita and M5, scenario linkage across datasets rather than a single shared enterprise system, intentionally interpretable methods, and pending Walmart and Rossmann robustness.

8 Conclusion

Forecast accuracy remains central to demand planning, but it is not sufficient for deployment. Operational systems execute planning signals, not forecast-error tables. This paper formulates forecast-driven demand planning as feasibility-constrained model selection and evaluates strategies using inventory exposure, planning volatility, switching burden, and execution-capacity violations alongside forecast accuracy.

The current results show that execution-risk-sensitive objectives can reorder deployable strategy rankings. They also show that the preferred deployable method depends on execution scenario, planning granularity, and demand intermittency. The manuscript remains extensible for additional

Walmart and Rossmann robustness experiments, richer execution-cost calibration, and enterprise workflow evidence.

A Appendix Snapshot Notes

This snapshot is intended for integrated reading of the current draft. Full literature citations are intentionally deferred until a verified citation pass is completed.